# The Effect of the Bologna Reform on the Graduate Labor Market

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# Motivation

- The Bologna reform is the biggest education reform of the 21$^{st}$ century in Europe
- Rare causal estimates of returns to higher education for middle-income countries
- No paper that evaluates the results of the Bologna reform in Russia

# Research question

- This paper estimates the effect of the Bologna reform on wages and employment in Russia

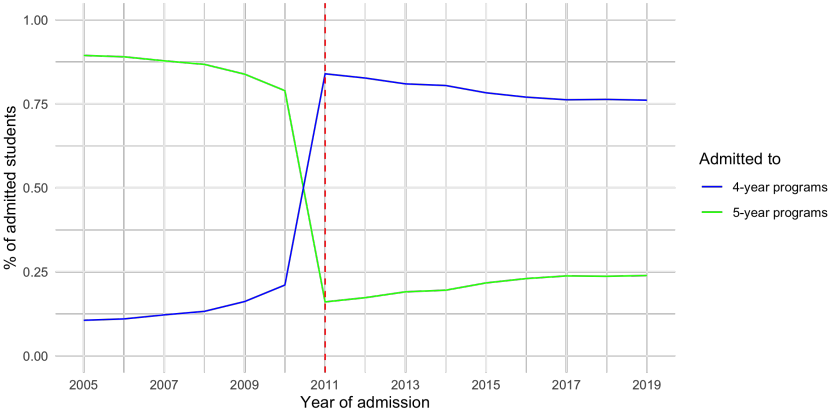
# Literature

- Bologna reform in European countries
  - Positive results of the reform on wages in Germany and Portugal and on employment in Italy (Bosio and Leonardi, 2011; Garra, 2013; Neugebauera and Weiss, 2018)
  - Negative impact on wages in Italy and on employment in Germany and Slovenia (Bosio and Leonardi, 2011; Farcnik and Domadenik, 2012; Neugebauera and Weiss, 2018)
- No paper about the results of the Bologna reform in Russia

# Institutional background

- 1992: 4-year programs were optional parallel to the 5-year degrees
- 2003: Russia signed the Bologna Declaration
- 2011: completed transformation from 5-year to 4-year programs

Figure: The share of 4– and 5–year full–time admitted students in public and private universities in 2005–2019




*Notes:* The red dashed line shows the year of the implementation of the reform.

# Data

- National Survey of Graduate Employment
- Repeated cross-sectional sample of 2010–2015 graduates
- 4– and 5–year university graduates
- 20–30 years old graduates who are not currently students
- Only full-time programs
- In total, there are 11,332 observations Descriptive statistics

# Empirical strategy

$$Y_{ipry} = \gamma_0 + \gamma_1 Treated_p + \gamma_2 Treated_p * Post_y + X_{ipr} + \lambda_p + \mu_y + \eta_r + \varepsilon_{ipry}$$

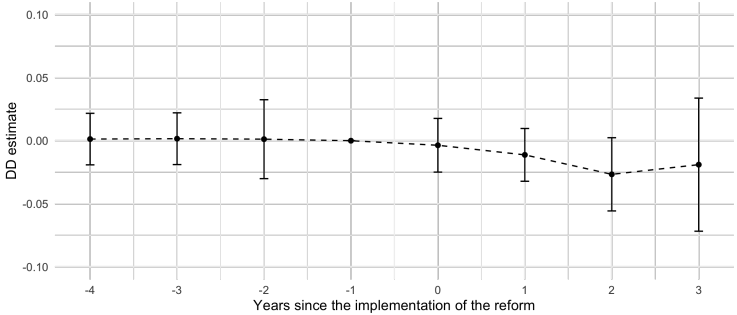
$Y_{ipr}$ is the labor market outcome of graduate $i$ from program $p$ in region $r$
$Treated_p$ is a dummy variable equal to 1 if a student has graduated from a 4–year program $p$ and 0 otherwise
$Treated_p * Post_y$ is the interaction of $Treated_p$ and a dummy variable for the year of graduation 2015
$X_{ipr}$ are exogenous individual characteristics (gender, type of institution (public/private), and funding (state–funded/fee–based))
$\lambda_p$ are program fixed effects
$\eta_r$ are region fixed effects
$\mu_y$ are the year of graduation fixed effects.

# Identification strategy

- Bologna reform offers a source of exogenous variation in the length of university education
- University applicants could not affect their decision to start 4– or 5–year programs
- The treatment is orthogonal to the potential outcomes, such as ability or self-selection

# Test for parallel pre–reform trends

Figure: Event-study estimates of a one–year reduction of higher education on employment




*Notes:* The figure shows the regression coefficients of interaction between the treatment variable and time period dummies with 95% confidence intervals. The omitted period is $-1$, the year before the implementation of the reform.

# OLS estimates

Table: The OLS estimates of the association between years of higher education and log monthly wages and employment

| | *Log monthly wages* | | | *Employed* | | |
|---|---|---|---|---|---|---|
| | All | Male | Female | All | Male | Female |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Years of education | 0.045** | 0.035* | 0.057** | 0.023** | 0.015 | 0.027** |
| | (0.013) | (0.014) | (0.017) | (0.007) | (0.010) | (0.011) |
| Pre–reform mean | 10.02 | 10.02 | 10.02 | 0.88 | 0.88 | 0.88 |
| Observations | 6,207 | 2,771 | 3,436 | 11,332 | 4,783 | 6,549 |

*Notes:* Robust standard errors clustered at the program and the graduation year level in parentheses.
*p<0.1; **p<0.05; ***p<0.01

# Main results

Table: The effect of a one–year reduction of higher education on log monthly wages, hourly wages, and employment for all

| | *Log monthly wages* | | *Log hourly wages* | | *Employed* | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Treated*Post | −0.010 | −0.018 | −0.021 | −0.026 | −0.014 | −0.015* |
| | (0.015) | (0.012) | (0.016) | (0.022) | (0.007) | (0.007) |
| Year and region FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Ind. controls | | ✓ | | ✓ | | ✓ |
| Pre–reform mean | 10.02 | 10.02 | 4.91 | 4.91 | 0.88 | 0.88 |
| Observations | 6,207 | 6,207 | 6,207 | 6,207 | 11,332 | 11,332 |

*Notes:* Robust standard errors clustered at the program and the graduation year level in parentheses.
$^{*}p<0.1$; $^{**}p<0.05$; $^{***}p<0.01$

# Heterogeneous results

Table: The effect of a one–year reduction of higher education on log monthly wages and employment separately by gender

| | *Log monthly wages* | | *Employed* | |
|---|---|---|---|---|
| | Male | Female | Male | Female |
| | (1) | (2) | (3) | (4) |
| Treated*Post | $-0.046^{**}$ | 0.017 | $-0.020$ | $-0.012$ |
| | (0.015) | (0.015) | (0.032) | (0.022) |
| Pre–reform mean | 10.13 | 9.92 | 0.95 | 0.82 |
| Observations | 2,771 | 3,436 | 4,783 | 6,549 |

*Notes:* Robust standard errors clustered at the program and the graduation year level in parentheses.

$^{*}$p<0.1; $^{**}$p<0.05; $^{***}$p<0.01

# Heterogeneous results

Table: The effect of a one–year reduction of higher education on log monthly wages and employment separately by funding

| | *Log monthly wages* | | *Employed* | |
|---|---|---|---|---|
| | State–funded | Fee–based | State–funded | Fee–based |
| | (1) | (2) | (3) | (4) |
| Treated*Post | −0.013 | −0.001 | −0.001 | −0.021 |
| | (0.020) | (0.010) | (0.015) | (0.015) |
| Pre–reform mean | 10.02 | 10.01 | 0.88 | 0.86 |
| Observations | 3,730 | 2,477 | 6,697 | 4,635 |

*Notes:* Robust standard errors clustered at the program and the graduation year level in parentheses.

$^{*}$p<0.1; $^{**}$p<0.05; $^{***}$p<0.01

# Mechanism

- As human capital theory suggests, shortening the length of university study period reduces the direct and indirect costs of obtaining a degree
  - the reduction in the length of study leads to the decreased productivity of a graduate, which will result in lower wages and unfavorable labor market opportunities
  - some students can compensate the reduction of the duration of their studies and invest in the accumulation of the human capital by studying more outside of the classroom

# Mechanism

Table: The effect of a one–year reduction of higher education on combining study and work and the level of on-the-job training

| | *Combining study and work* | | *On-the-job training* | |
|---|---|---|---|---|
| | Male | Female | Male | Female |
| | (1) | (2) | (3) | (4) |
| Treated*Post | $0.081^{***}$ | $-0.042^{**}$ | $0.072^{**}$ | $-0.045^{**}$ |
| | (0.015) | (0.012) | (0.024) | (0.013) |
| Pre–reform mean | 0.28 | 0.20 | 0.14 | 0.13 |
| Observations | 4,783 | 6,549 | 1,385 | 1,747 |

*Notes:* Robust standard errors clustered at the program and the graduation year level in parentheses.

$^{*}p<0.1$; $^{**}p<0.05$; $^{***}p<0.01$

# Possible threats to the identification strategy

- Changes in the cohort composition Share of applicants Number of applicants Characteristics of students Share of graduates
- An introduction of the standardised exam for admission to universities Characteristics of applicants
- The 2015 Russian financial crisis Employment in 2014–2018

# Concluding remarks

- The Bologna reform offers exogenous variation in the length of university education
- The reform did not harm the employment of graduates
- No adverse effect of a 20% reduction in the length of higher education on wages for graduates who spent more time studying
  - negative effects for students who worked during their studies
- I provide suggestive evidence for the compensation–based mechanism to explain these results
- Long-term effects are to be estimated

Table: Summary statistics

| Variable | Observations | Mean | SD | Min | Max |
|---|---|---|---|---|---|
| *4–year university graduates* | | | | | |
| Male | 1,722 | 0.41 | 0.49 | 0 | 1 |
| Public institution | 1,722 | 0.94 | 0.23 | 0 | 1 |
| State–funded | 1,722 | 0.53 | 0.50 | 0 | 1 |
| Employed | 1,722 | 0.82 | 0.38 | 0 | 1 |
| Monthly wage, RUB | 887 | 23,279 | 11,092 | 5,000 | 130,000 |
| *5–year university graduates* | | | | | |
| Male | 9,610 | 0.42 | 0.49 | 0 | 1 |
| Public institution | 9,610 | 0.95 | 0.21 | 0 | 1 |
| State–funded | 9,610 | 0.60 | 0.49 | 0 | 1 |
| Employed | 9,610 | 0.86 | 0.35 | 0 | 1 |
| Monthly wage, RUB | 5,320 | 24,539 | 12,009 | 5,000 | 120,000 |

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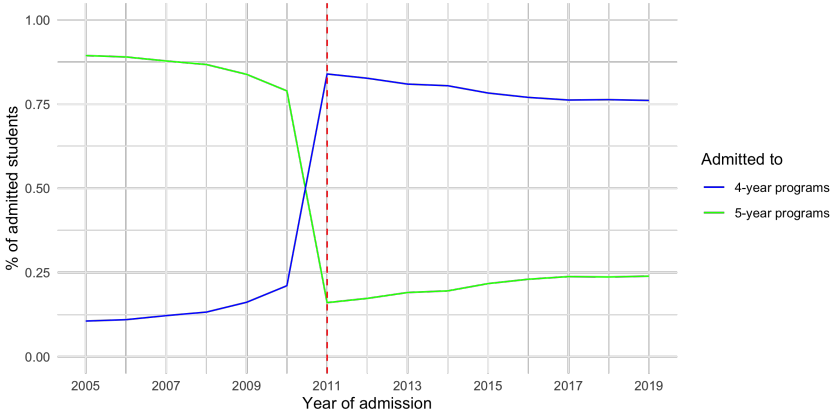
Figure: The share of 4– and 5–year full–time admitted students in public and private universities in 2005–2019

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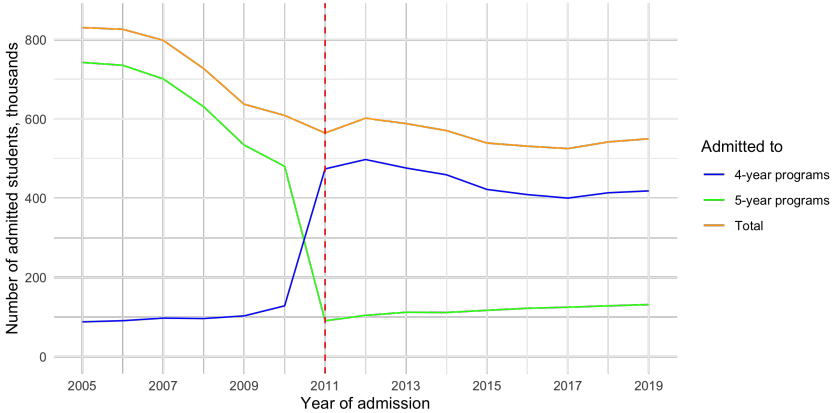
*Notes:* The red dashed line shows the year of the implementation of the reform.

Figure: The number of 4– and 5–year full–time admitted students in public and private HEIs in 2005–2019, thousands

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*Notes:* The red dashed line shows the year of the implementation of the reform.

Table: The characteristics of students by the year of graduation

| Variable | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 |
|---|---|---|---|---|---|---|
| Male | 0.44 | 0.44 | 0.45 | 0.42 | 0.40 | 0.39 |
| | (0.50) | (0.50) | (0.50) | (0.49) | (0.49) | (0.49) |
| Public institution | 0.94 | 0.96 | 0.95 | 0.95 | 0.96 | 0.95 |
| | (0.23) | (0.20) | (0.22) | (0.22) | (0.19) | (0.21) |
| State–funded | 0.59 | 0.60 | 0.59 | 0.59 | 0.59 | 0.58 |
| | (0.49) | (0.49) | (0.49) | (0.49) | (0.49) | (0.49) |

*Notes:* This table shows sample means and standard deviations from the the National Survey of Graduate Employment carried out by the Russian Federal State Statistics Service.

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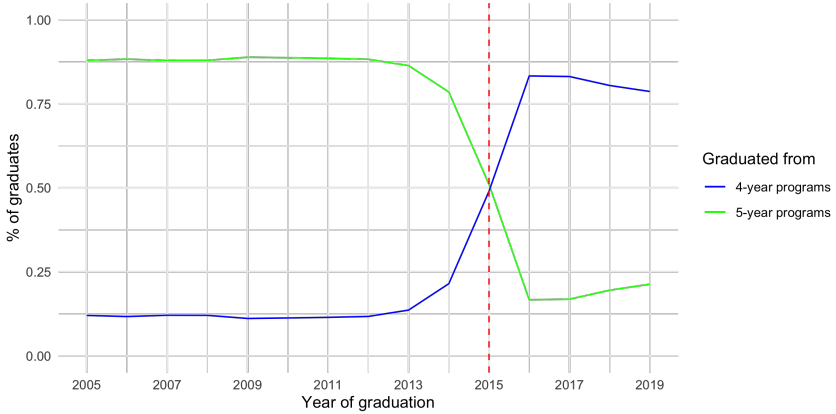
Figure: The share of 4– and 5–year full–time graduates in public and private universities in 2005–2019




*Notes:* The red dashed line shows the year of a double flow of graduates.

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Table: The characteristics of graduates when they were admitted to university

| Variable | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 |
|---|---|---|---|---|---|---|
| Age at the year of application | 17.38 | 17.45 | 17.62 | 17.66 | 17.82 | 17.73 |
| | (1.09) | (1.21) | (1.26) | (1.27) | (1.39) | (1.35) |
| Rural residence | 0.21 | 0.23 | 0.24 | 0.28 | 0.27 | 0.28 |
| | (0.41) | (0.42) | (0.43) | (0.45) | (0.44) | (0.45) |
| Moscow and Saint Petersburg | 0.13 | 0.13 | 0.13 | 0.12 | 0.10 | 0.12 |
| | (0.34) | (0.33) | (0.34) | (0.32) | (0.30) | (0.33) |

*Notes:* This table shows sample means and standard deviations from the the National Survey of Graduate Employment carried out by the Russian Federal State Statistics Service.

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